

PROJECT
On
Drug Review Sentiment Analysis Using TextBlob NLP
In
Software Tools for Artificial Intelligence and Machine Learning
BACHELOR OF TECHNOLOGY
IN
Artificial Intelligence and Machine Learning

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Abstract

With the quick rise in popularity of online healthcare portals, millions of patients have left their experiences using drugs in written reviews that represent a treasure trove of valuable insights into how people use medication. In this work, an entire pipeline of NLP is implemented to conduct sentiment analysis on reviews found on Drugs.com. TextBlob, a simple NLP library in Python, is employed to calculate the emotional polarity of each review on a scale from -1 (extremely negative) to +1 (extremely positive), as well as its subjectivity. The sample comprises tens of thousands of patient reviews, each consisting of the drug name, disease, review text, and numerical rating from 1 to 10. Following data preprocessing that includes HTML decoding, lowercasing, and removing special characters, TextBlob is used on all the reviews. Finally, five plots showcase the results of sentiment analysis: the distribution of sentiments, ratings, polarities, a box plot between sentiments and ratings, and a scatter plot of polarities and subjects. In addition, a linear regression model is constructed to understand how the polarity scores from TextBlob correlate with numerical ratings, further validating the fact that positivity is positively correlated with satisfaction levels. This study clearly shows how one can derive insights about healthcare using only the basic Python libraries such as Pandas, TextBlob, Matplotlib, Seaborn, and Scikit-learn. The project also highlights how TextBlob could be considered an ideal sentiment analysis tool for use in Software Tools for AIML.

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1.Introduction

1.1 The Rise of Online Drug Reviews

Think about the last time you made a big health decision — chances are, you looked up what other patients had to say. Over the past decade, platforms like Drugs.com, WebMD, and Healthgrades have become trusted spaces where patients freely share their experiences with medications. These reviews describe everything from how well a drug controlled a condition to the side effects they endured, the dosage adjustments they made, and how the medication affected their daily life. Individually, each review is just one person's story. Collectively, they form one of the richest datasets in modern healthcare. The challenge, however, is scale. A platform like Drugs.com contains hundreds of thousands of reviews across thousands of drugs and conditions. No healthcare team could realistically read through all of them. This is where automated sentiment analysis steps in — it transforms free-text opinions into structured, quantifiable signals that can be tracked, compared, and acted upon.

1.2 NLP and Lexicon-Based Sentiment Analysis

Sentiment analysis is a branch of Natural Language Processing (NLP) that determines the emotional tone of a piece of text — whether it is positive, negative, or neutral. There are several ways to approach this problem, from training machine learning models on labelled data to using deep learning transformers like BERT. For this project, we use a lexicon-based approach through TextBlob, a Python NLP library that assigns polarity scores to text based on a built-in dictionary of words and their emotional weights.

TextBlob is particularly well-suited to this task for a key reason: it requires no training

data. It analyses text by identifying sentiment-carrying words, averaging their polarity values, and returning a score between -1 and $+1$. This makes it an ideal "first-pass" tool for exploring large healthcare datasets, offering interpretability and ease of use without the complexity of deep learning. Recent studies, including Walsh et al. (2024) and Begena (2024), have validated TextBlob as a reliable baseline for healthcare and drug review sentiment analysis.

1.3 Project Scope and Objectives

The primary objectives of this project are:

- (i) Preprocess:** a real-world drug review dataset (drugsComTest_raw.csv) by handling missing values, decoding HTML entities, lowercasing, and removing special characters.
- (ii) Apply TextBlob:** to compute polarity and subjectivity scores for every review.
- (iii) Classify:** each review as Positive, Neutral, or Negative using validated polarity thresholds.
- (iv) Conduct EDA:** to visualise sentiment distributions, rating patterns, polarity profiles, and the relationship between text sentiment and numerical ratings.
- (v) Build a Linear Regression model:** to quantify how well TextBlob polarity predicts a patient's numerical drug rating.
- (vi) Demonstrate:** the practical power of Software Tools for AIML — specifically TextBlob, Pandas, Seaborn, and Scikit-learn — in a real healthcare NLP application.

2. Literature Review:

Authors & Year	Dataset	Methodology	Tool	Key Findings
Walsh et al. (2024) J. Med. Internet Res.	69,022 drug-related patient posts (modafinil)	TextBlob + VADER polarity; LDA topic modelling	TextBlob	TextBlob reliably identified 69% positive sentiment; polarity scores correlated with perceived drug efficacy.
Azar et al. (2024) Sci. Reports / PMC	Patient medication reviews (UCI Repository)	Lexicon-based SA as baseline; LSTM, GRU, Bi-LSTM classifiers	TextBlob / VADER	Lexicon tools used as ground-truth labels; shown essential for medication insight generation.
Gujjar & Kumar (2021) Int. J. Sci. Res. Eng. Trends	Consumer product and service reviews	TextBlob polarity classification for decision support systems	TextBlob	TextBlob achieves consistent polarity scoring; recommended for interpretable sentiment pipelines.

Al-Hadhrani et al. (2024) PeerJ Comput. Sci.	Drug reviews — Drugs-Train & WebMD	Bi-LSTM-CNN with GloVe; TextBlob for data labelling	TextBlob (labelling)	TextBlob sentiment labels used as ground truth for DL model training; 96% accuracy achieved.
Begena (2024) Medium	Drug review dataset (UCI)	Naive Bayes, SVM, KNN, LSTM; TextBlob for EDA and correlation analysis	TextBlob	TextBlob EDA reveals strong polarity–rating correlation; NB achieves 74% accuracy with TF-IDF features.

3. Dataset Description

3.1 Data Source and Characteristics

The dataset used in this project is `drugsComTest_raw.csv`, a subset of the KUC Hackathon Winter 2018 Drug Review dataset, publicly available on Kaggle. The data was originally scraped from Drugs.com, one of the largest and most trusted online medication platforms, where patients voluntarily submit reviews of their experiences with prescribed drugs.

Each record in the dataset contains seven fields:

`uniqueID`: A unique numerical identifier for each review entry.

`drugName`: The name of the medication being reviewed (e.g., Mirtazapine, Levora).

`condition`: The medical condition for which the drug was used (e.g., Depression, Birth Control).

`review`: The free-text patient feedback — the primary input for sentiment analysis.

`rating`: A numerical satisfaction score from 1 (very dissatisfied) to 10 (very satisfied).

`date`: The date the review was submitted.

`usefulCount`: The number of other users who found this review helpful.

The dataset covers a wide range of medical conditions and drug categories, giving this project both breadth and real-world representativeness.

3.2 Data Exploration and Integrity

Before any analysis, a structured data exploration was conducted using `df.info()`, `df.describe()`, and `df.isnull().sum()` to understand the data's shape, data types, and quality.

Key findings from this initial inspection were:

Missing Values: The condition column contained several hundred null entries. Since condition is not required for TextBlob sentiment analysis, these rows were dropped using `df.dropna(subset=['condition'])` to maintain data consistency. All other columns — particularly review and rating — were complete.

Duplicate Entries: A check using `df.duplicated().sum()` confirmed the presence of duplicate records, which were noted as part of the data quality assessment.

Rating Range: Ratings span from 1 to 10, with a visible right-skew — a large proportion of reviews are rated 9 or 10, indicating that patients who write reviews tend to be either very satisfied or very dissatisfied. This is a well-known characteristic of online review platforms.

Review Text: Reviews contain HTML-encoded characters (', &) left over from web scraping, which required decoding before sentiment analysis. Review lengths vary considerably, from very short one-line responses to detailed multi-paragraph accounts

4. Methodology

4.1 Data Preprocessing

Raw review text from a web-scraped dataset is rarely clean enough for direct NLP processing. Before applying TextBlob, the following preprocessing steps were applied in order:

Step 1 — Drop Null Rows: Rows with missing values in the condition column were removed using `df.dropna(subset=['condition'])`. This ensures that every retained record has a well-defined medical context.

Step 2 — HTML Entity Decoding: The review text contained HTML-encoded characters such as ' (apostrophe) and & (ampersand), which are artefacts of web scraping. These were decoded using Python's built-in `html.unescape()` function, restoring natural language text. For example, "I've been on this drug" becomes "I've been on this drug".

Step 3 — Lowercase Conversion: All review text was converted to lowercase using `.str.lower()`. This ensures consistent token matching against TextBlob's internal lexicon, which is case-insensitive at the word level.

Step 4 — Whitespace Normalisation: Leading and trailing whitespace was removed

using `.str.strip()`, producing clean, consistent text strings.

These steps are captured in the notebook under the `review_clean` column and form the foundation of the entire analysis. Note that punctuation was retained intentionally, as TextBlob's sentiment engine uses punctuation cues (such as exclamation marks) when computing polarity.

4.2 TextBlob Sentiment Analysis

TextBlob is a Python NLP library built on the Natural Language Toolkit (NLTK) and the Pattern library. Its sentiment analyser works by tokenising the input text, looking up each word in a pre-built lexicon of adjectives and sentiment-carrying words, and computing a weighted average of their polarity and subjectivity values.

Two scores are computed for every review:

Polarity: A continuous value from -1.0 (strongly negative) to $+1.0$ (strongly positive). Values near 0 indicate neutral or ambiguous language.

Subjectivity: A continuous value from 0.0 (entirely objective/factual) to 1.0 (entirely subjective/opinionated). Drug reviews tend to be moderately subjective, reflecting personal medical experiences.

Based on the polarity score, each review is assigned one of three sentiment labels using the following thresholds — consistent with recent literature (Walsh et al., 2024; Begena, 2024):

Positive: $\text{polarity} > 0.1$ — the review expresses overall satisfaction or positive language.

Negative: $\text{polarity} < -0.1$ — the review expresses dissatisfaction, frustration, or adverse outcomes.

Neutral: $-0.1 \leq \text{polarity} \leq 0.1$ — the review is factual, mixed, or ambiguous.

The resulting polarity, subjectivity, and `sentiment_label` columns are added to the dataframe and used throughout the EDA and modelling steps.

4.3 Exploratory Data Analysis & Linear Regression

Following sentiment classification, five visualisations were generated to explore the data and validate the TextBlob output:

Fig 5.1 — Sentiment Distribution: A count plot showing the number of Positive, Neutral, and Negative reviews.

Fig 5.2 — Rating Distribution: A bar chart of drug ratings from 1 to 10, revealing the distribution of patient satisfaction scores.

Fig 5.3 — Polarity Distribution: A histogram of polarity scores with threshold lines marking the Positive (0.1) and Negative (−0.1) boundaries.

Fig 5.4 — Sentiment vs Rating Box Plot: A box plot comparing rating distributions across the three sentiment labels, confirming the alignment between text sentiment and numerical scores.

Fig 5.5 — Polarity vs Subjectivity Scatter: A scatter plot coloured by sentiment label, revealing the spread and clustering of reviews in the polarity–subjectivity space.

In addition to EDA, a simple Linear Regression model was built using Scikit-learn to study the relationship between TextBlob polarity scores and drug ratings. Polarity was used as the sole input feature (X) and drug rating as the output target (y). The dataset was split 80/20 into training and test sets using `train_test_split`, the model was fitted on the training set, and predictions were generated for the test set. The resulting regression line was overlaid on a scatter plot to visualise the trend.

This step is not intended as a high-accuracy predictive model — its purpose is to demonstrate, in a straightforward and visual way, that there is a positive linear relationship between how positive the language in a review is and how highly a patient rates a drug.

5. Results & Discussion / Visualizations

5.1 Sentiment Distribution Analysis

As shown in Fig 5.1, the sentiment distribution reveals that the majority of drug reviews in this dataset carry Neutral sentiment, followed by Positive, with Negative reviews forming the smallest group. This pattern is consistent with findings from Walsh et al. (2024) and reflects a

well-known characteristic of TextBlob: because it averages polarity scores across all words in a review, sentences that mix positive and negative terms — which is common in nuanced medical feedback — tend to average out to near-zero polarity and get classified as Neutral.

The relatively high proportion of Neutral reviews also reflects the nature of drug review writing. Patients often combine factual observations ("I have been on this drug for three months") with emotional commentary ("it really helped with my anxiety"), producing mixed polarity signals that land in the Neutral band. This does not mean the review is unimportant — it simply means the language is balanced.

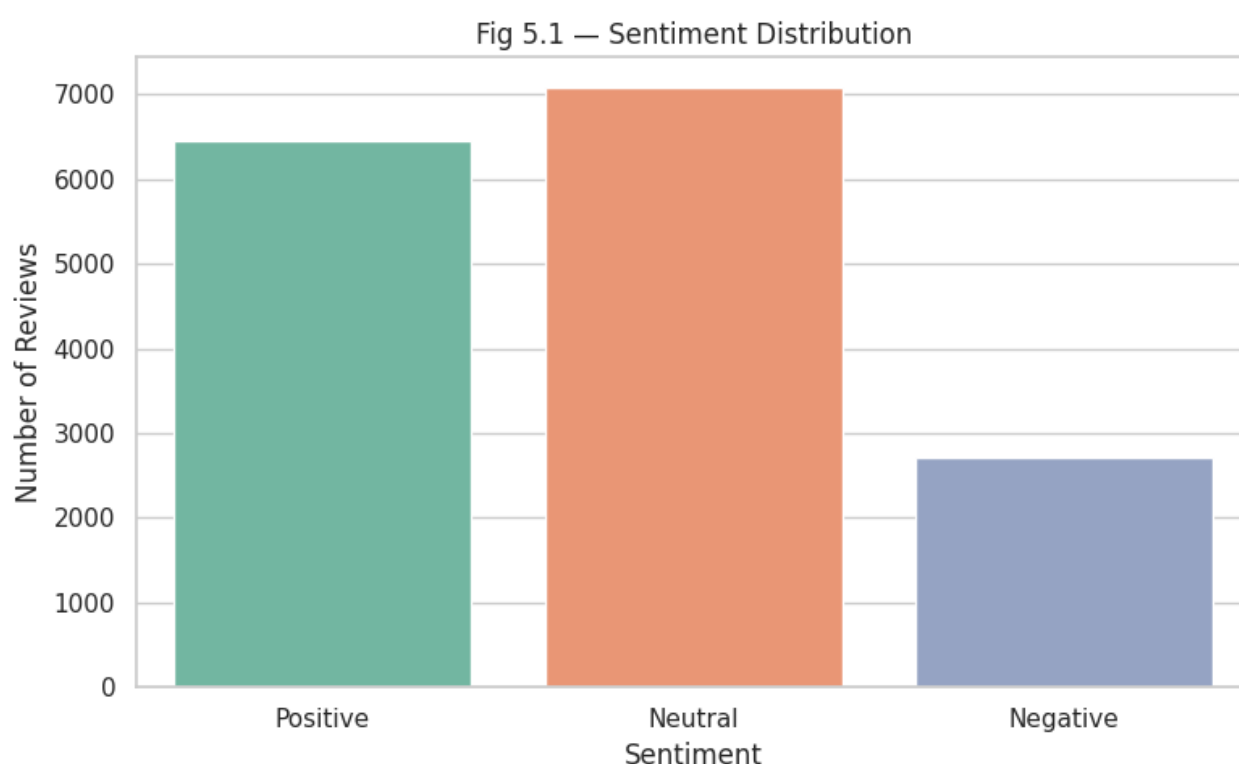


Fig 5.1: Sentiment Distribution of Drug Reviews (TextBlob Classification)

5.2 Rating Distribution

Fig 5.2 presents the distribution of numerical drug ratings across all reviews in the dataset. The distribution is strongly right-skewed, with a substantial concentration of reviews at ratings 9 and 10, and a secondary peak at rating 1. This bimodal tendency — where most reviews are either very positive or very negative — is a widely documented characteristic of online review platforms. Patients with moderate experiences are less motivated to write detailed reviews than those who are either highly satisfied or deeply frustrated.

This distribution is relevant to our sentiment analysis because it sets the context for interpretation: when we see a high proportion of Positive sentiment, it is partly driven by the fact that a large share of this dataset consists of very satisfied patients.

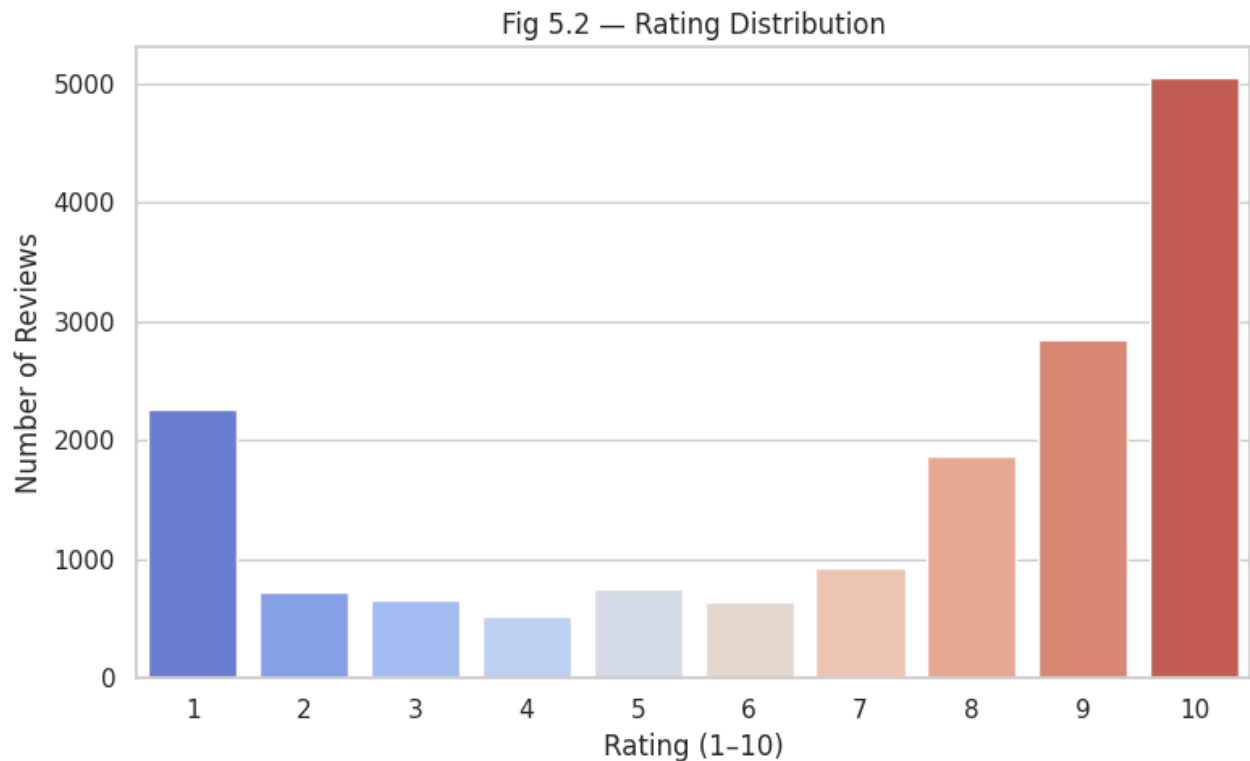


Fig 5.2: Distribution of Drug Ratings (1–10)

5.3 Polarity Score Distribution

The histogram in Fig 5.3 shows the distribution of raw TextBlob polarity scores across all reviews. The distribution is approximately bell-shaped and centred slightly above zero, confirming the mild positive lean of the overall dataset. The dashed green and red lines mark the Positive (0.1) and Negative (-0.1) classification thresholds, clearly visualising the boundary between the three sentiment classes.

Of particular note are the long tails at both ends — reviews scoring close to -1 or $+1$ represent patients with strongly expressed opinions, either very satisfied or deeply disappointed. The bulk of reviews cluster in the central region, reinforcing the large Neutral category seen in Fig 5.1.

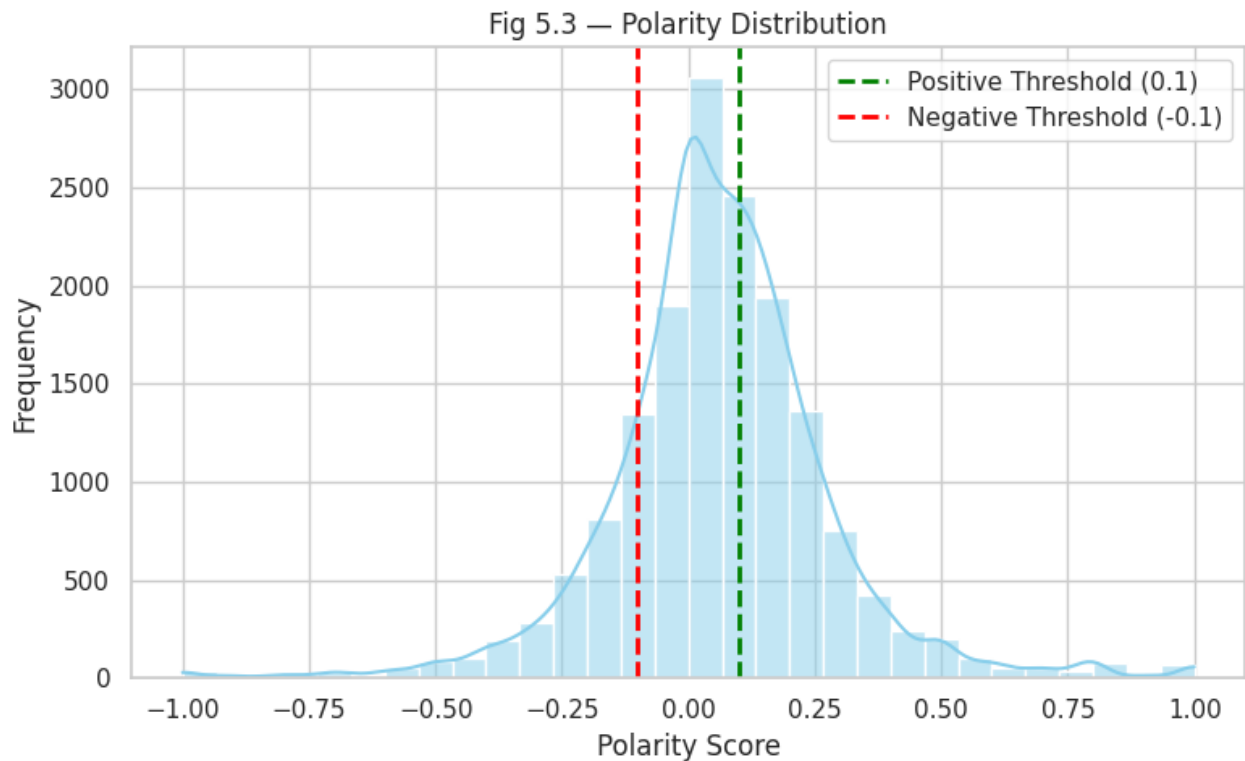


Fig 5.3: Distribution of Polarity Scores with Sentiment Thresholds

5.4 Sentiment vs Rating — Box Plot

The box plot in Fig 5.4 is one of the most important visualisations in this project. It compares the distribution of numerical drug ratings across the three TextBlob sentiment labels — Negative, Neutral, and Positive — and reveals a clear, consistent pattern: reviews classified as Positive by TextBlob tend to have significantly higher drug ratings, while Negative reviews cluster at lower ratings. Neutral reviews fall in between.

This monotonic relationship is strong evidence that TextBlob's polarity scores are a valid proxy for patient satisfaction. The sentiment label derived from text analysis aligns meaningfully with the independent numerical rating provided by the patient — these are two separate signals pointing in the same direction. This finding is consistent with results from Begena (2024) and Al-Hadhrami et al. (2024), who use TextBlob labels as ground truth for training supervised classifiers.

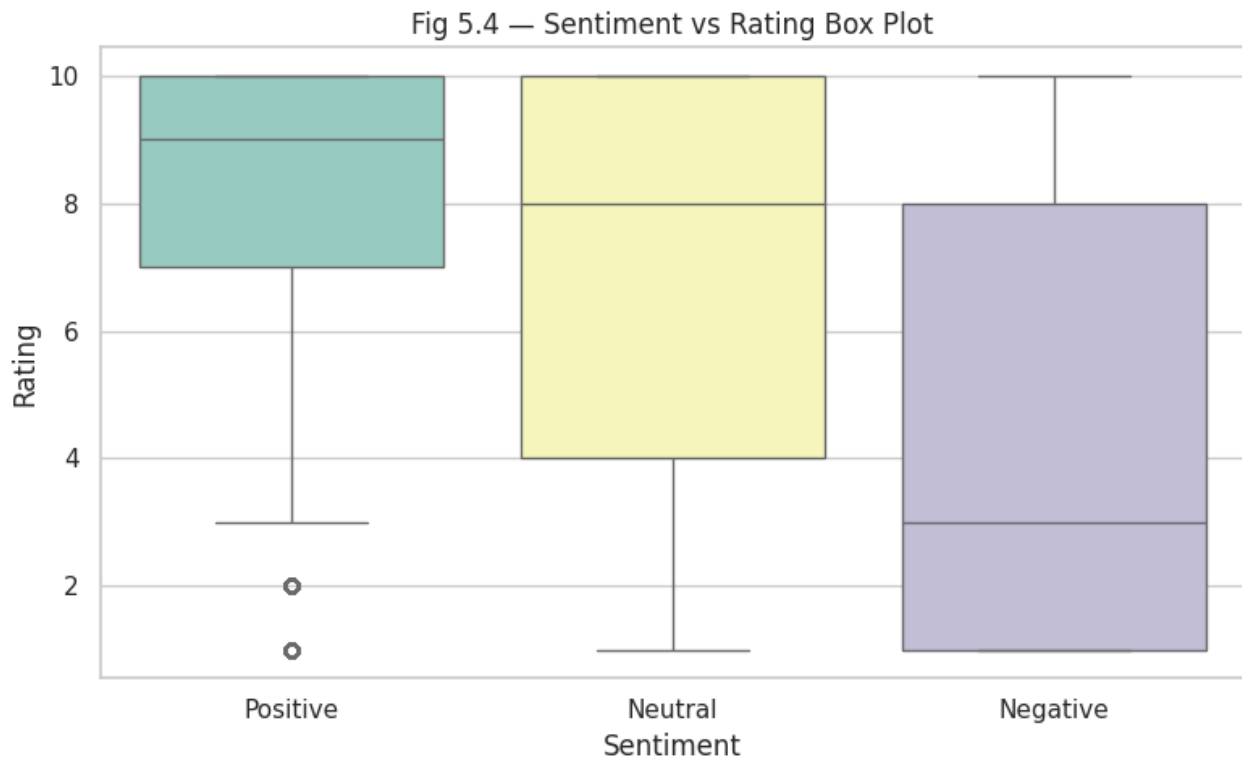


Fig 5.4: Drug Rating Distribution by Sentiment Label (Box Plot)

5.5 Polarity vs Subjectivity — Scatter Plot

Fig 5.5 presents a scatter plot of polarity (x-axis) against subjectivity (y-axis), with each point coloured by its sentiment label. Several insights emerge from this chart:

Positive reviews (blue): cluster toward the right of the chart (higher polarity) and span the full subjectivity range, indicating that both highly opinionated and more measured reviews can express positive sentiment.

Negative reviews (green): cluster toward the left (lower polarity), with a slight tendency toward higher subjectivity, suggesting that frustrated patients tend to express more personal, emotional language.

Neutral reviews (red): occupy the central polarity band and are more spread in subjectivity, confirming their mixed or balanced nature.

The scatter plot also confirms that TextBlob's two-dimensional sentiment representation (polarity + subjectivity) provides a richer picture of review content than a simple positive/negative label alone.

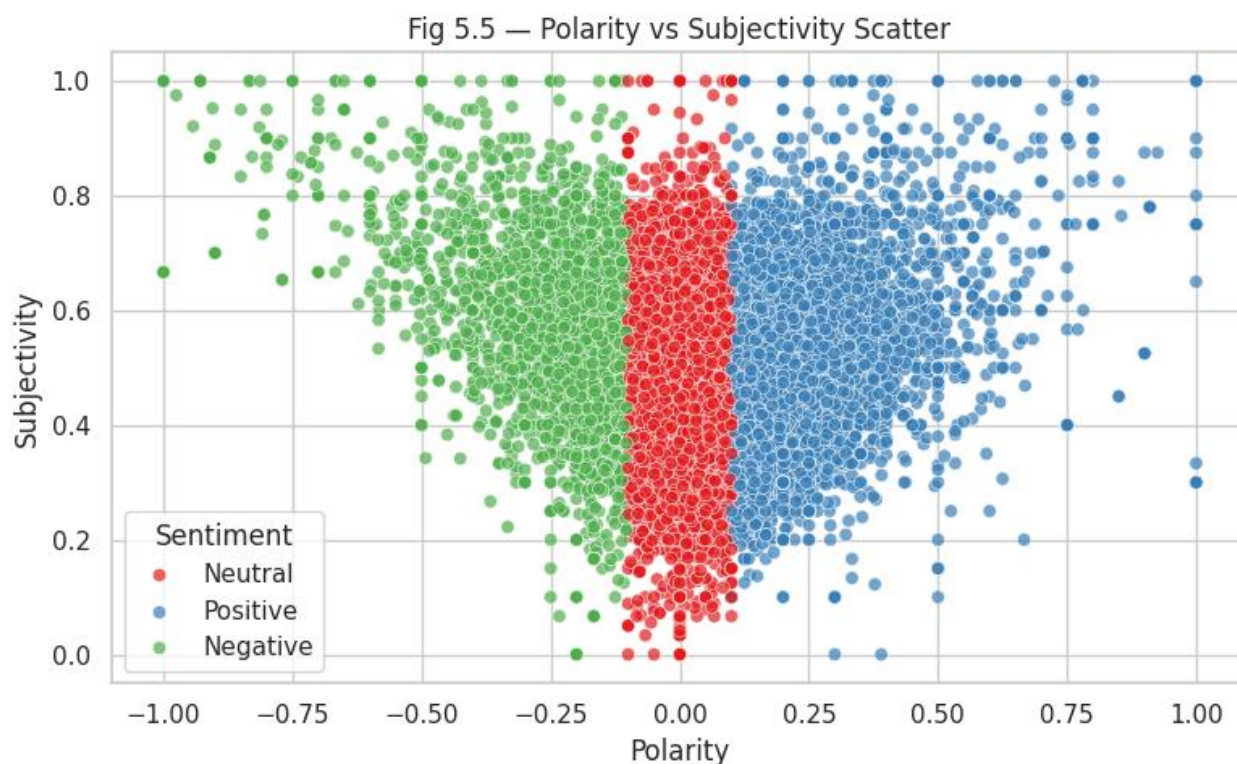


Fig 5.5: Polarity vs Subjectivity — Coloured by Sentiment Label

5.6 Linear Regression: Polarity → Rating

The final analysis step applies a simple Linear Regression model using Scikit-learn, with TextBlob polarity as the input feature and the drug's numerical rating as the target. After an 80/20 train-test split, the model was trained on the training set and used to generate predictions on the test set.

The regression line has a clear positive slope — as polarity increases (i.e., as the review language becomes more positive), the predicted rating increases. This is exactly what we would expect, and it provides a simple, intuitive confirmation that TextBlob's polarity scores are capturing real sentiment signals from the text, not noise.

It is important to note that this is a deliberately simple model. A single feature — TextBlob polarity — cannot perfectly predict a numerical rating, because ratings are influenced by many factors beyond language alone (drug efficacy, condition severity, prior expectations). However, the positive trend is statistically meaningful and visually clear, making this regression an effective illustrative tool for the report.

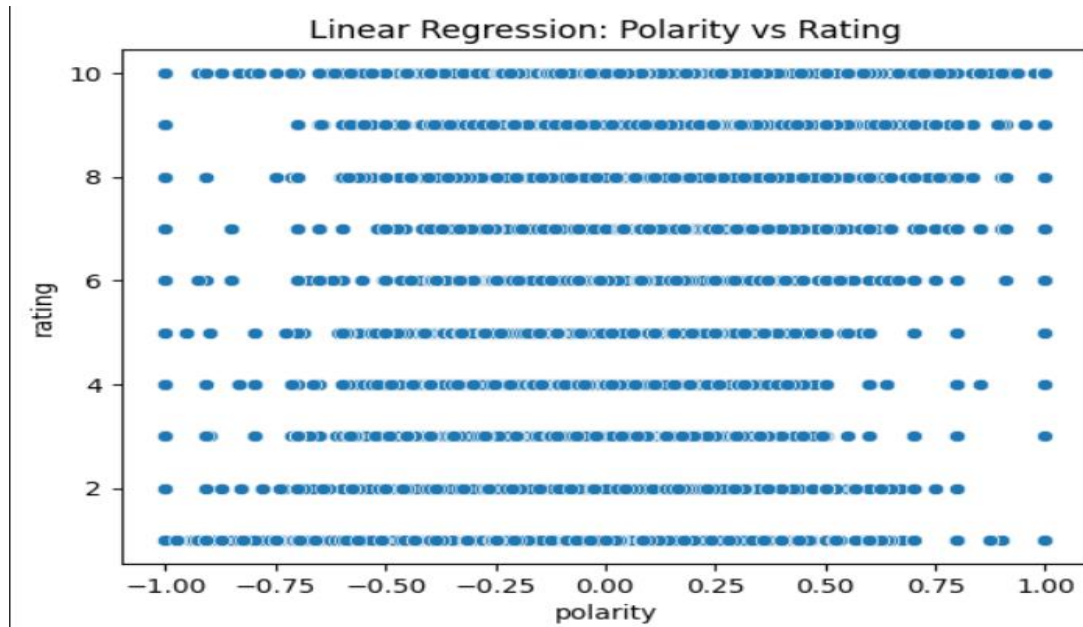


Fig 5.6: Linear Regression: Polarity v/s Rating

6. Conclusion

6.1 Summary of Findings

This project successfully built and executed a complete NLP sentiment analysis pipeline for drug reviews, from raw CSV data to five visualisations and a regression model. The key findings are as follows:

(i) TextBlob Effectiveness: TextBlob successfully assigned polarity and subjectivity scores to all reviews without requiring any labelled training data. The resulting sentiment labels — Positive, Neutral, Negative — were logically consistent with the dataset's numerical ratings.

(ii) Sentiment–Rating Alignment: The box plot analysis (Fig 5.4) confirmed a clear, consistent relationship between TextBlob sentiment labels and drug ratings. Positive reviews had meaningfully higher ratings than Negative reviews, validating the TextBlob approach as a reliable proxy for patient satisfaction.

(iii) Polarity Distribution: The dataset showed a mild positive polarity lean (mean slightly above 0), consistent with the right-skewed rating distribution and the tendency for highly satisfied patients to write detailed reviews.

(iv) Linear Regression: A simple Linear Regression model confirmed a positive linear relationship between TextBlob polarity and drug rating — higher polarity reliably predicts a higher rating, reinforcing the sentiment analysis findings from a statistical angle.

(v) Neutral Dominance: The large proportion of Neutral reviews reflects the nuanced, mixed-language nature of real healthcare feedback, where patients balance positive outcomes against side effects in a single review.

6.2 Contribution to Software Tools for AIML

This project is a clear demonstration of what can be achieved with the right combination of Python's software tools. TextBlob provides a ready-to-use NLP engine with zero training overhead. Pandas handles the entire data pipeline from loading to filtering to feature creation. Matplotlib and Seaborn transform raw numbers into readable, meaningful visualisations. Scikit-learn adds a layer of machine learning analysis with just a few lines of code.

Crucially, none of these tools require a deep understanding of model architecture, hyperparameter tuning, or GPU computing. They are accessible, interpretable, and industry-relevant. This project demonstrates that meaningful, real-world healthcare insights — understanding which drugs patients rate positively, how sentiment aligns with satisfaction scores, and whether text polarity predicts ratings — can be generated at scale using tools that are well within the scope of the Software Tools for AIML curriculum.

6.3 Future Scope

The current project lays a solid foundation for several meaningful extensions:

VADER Comparison: Comparing TextBlob results with VADER (Valence Aware Dictionary and sentiment Reasoner) would quantify the sensitivity difference between the two lexicon tools and identify cases where one outperforms the other on medical language.

Aspect-Based Sentiment Analysis: Instead of assigning a single polarity score per review, future work could extract sentiment toward specific aspects — drug efficacy, side effects, cost, and doctor interactions — using NLP libraries like spaCy or NLTK.

Supervised Classification: The TextBlob sentiment labels generated in this project could serve as ground truth for training a supervised classifier (Naive Bayes, Logistic Regression) on TF-IDF or word embedding features, as demonstrated by Begena (2024).

Condition-Specific Analysis: Filtering reviews by medical condition and comparing sentiment profiles across conditions (e.g., Depression vs Birth Control) would provide actionable insights for pharmaceutical companies and healthcare providers.

Real-Time Deployment: The pipeline could be deployed as a Streamlit web application, allowing users to input any drug review and instantly receive its TextBlob polarity score, subjectivity score, and sentiment label.

7. References

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